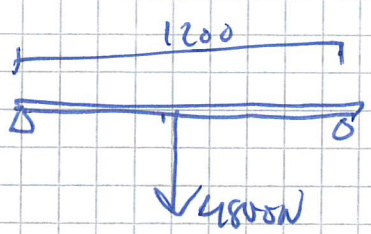
67



$R_e = 320\ N/mm^2$

$n_s = ?$

$$\frac{R_e}{n_s} = \frac{M_b}{W_b} \Rightarrow n_s = \frac{R_e W_b}{M_b} = \frac{R_e \cdot \pi \cdot d^3 \cdot 4}{32 \cdot F \cdot l} = 3{,}2$$

$$3{,}2 = \frac{320 \cdot \pi \cdot d^3 \cdot 4}{32 \cdot 4800 \cdot 1200} \Rightarrow d = \sqrt[3]{\frac{3{,}2 \cdot 32 \cdot 4800 \cdot 1200}{320 \cdot \pi \cdot 4}} =$$

$$= \underline{52{,}73\ mm}$$

78. $d = 60\,mm$

$M_v = 6\,kNm$

$$\tau_{max} = \frac{M_v}{W_v} = \frac{6000000}{\frac{\pi \cdot 60^3}{16}} = 141\ N/mm^2$$

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79. $D = 60\,mm$

$d = 40\,mm$

$$W_v = \frac{\pi(D^4 - d^4)}{16D} = \frac{\pi(60^4 - 40^4)}{16 \cdot 60} =$$

$$\tau_{max} =$$

---

80. $D = 40\,mm$

$$\tau_{max} = \frac{2400000}{\frac{\pi \cdot 40^3}{16}} =$$

81.

$$\tau_{skut} = \frac{1400000}{\frac{\pi(50^4 - 35^4)}{16 \cdot 50}} =$$

50667606

82.

$M_v = 300 \cdot F$

$\sigma_{tah} = 420 N/mm^2$

$\tau_{dov} = 0{,}6 \cdot 420 = 252$

$n_s = 1{,}5$

$D = 40 mm$

$$\tau_{dov} = \frac{300F \cdot 16}{\pi D^3} \Rightarrow F = \frac{\tau_{dov} \cdot \pi \cdot D^3}{300 \cdot 16} =$$

$$= \frac{252 \cdot \pi \cdot 40^3}{300 \cdot 16} = 10555 N \qquad \frac{10533}{1{,}5} = 7037 N \Rightarrow 700 kg$$

84823

83. $M_v = 400000\ Nmm$
$n_s = 1,6$
$D = 30mm$

$$\tau_{tin} = \frac{400000 \cdot 16}{\pi \cdot 30^3} = 75,45$$

$$\sigma_{tu} \frac{\tau_{tin} \cdot n_s}{0,6} = \frac{75,45 \cdot 1,6}{0,6} = 201\ N/mm^2$$

---

84 $n_s = ?$

42875
134695

$M_v = 900000\ Nmm$

$\sigma_{tu} = 400\ N/mm^2$

$D = 35mm$

$$\tau_{tin} = \frac{900000 \cdot 16}{\pi \cdot 35^3} = 106\ N/mm^2$$

σ_{tu}

$$\frac{\tau_{tin}}{0,6} = 178\ N/mm^2 \quad n_s = \frac{400}{178} = 2,24$$

$\tau_{tn/r}$

---

85. $D = 30mm$
$n = 600\ rpm$
$\sigma_{tin} = 480\ N/mm^2$
$n_s = 1,5$

$$\frac{0,6 R_e}{n_s} = \frac{M_v}{W_v} \Rightarrow M_v = \frac{0,6 R_e \cdot W_v}{n_s} =$$

$$= \frac{0,6 \cdot 460 \cdot \pi \cdot 30^3}{16 \cdot 1,5} = 975464\ Nmm$$

obs Nm

$$M_v = 9,55 \cdot \frac{P}{n} \Rightarrow P = \frac{M_v \cdot n}{9,55} = \frac{975,464 \cdot 600}{9,55} = 61kW$$

86.

$\sigma_{fü} = ?$

$D = 25\,mm$

$P = 20000\,W$

$n = 480\,rpm$

$n_s = 1{,}4$

$$M_v = 9{,}55 \cdot \frac{20000}{480} = 397{,}9\,Nm$$

$$W_v = \frac{\pi \cdot 25^3}{16} = 3068\,mm^3$$

$\tau_{fü} = 129{,}7$ $\quad \sigma = \frac{129{,}7}{0{,}6} = 216{,}15$ $\quad 216{,}15 \cdot 1{,}4 = 302{,}6\,N/mm^2$

---

87. $P = 30000\,W$ $\quad R_e = 350\,N/mm^2$

$n = 420\,rpm$

$\sigma_{fü} = 350\,N/mm^2$

$n_s = 2$

$D = ?$

$$0{,}6\frac{R_e}{n_s} = \frac{M_v}{W_v} \Rightarrow \frac{0{,}6R_e}{n_s} = \frac{9{,}55 \cdot \frac{P}{n}}{\frac{\pi \cdot D^3}{16}} \Rightarrow$$

$$\Rightarrow \frac{0{,}6R_e}{n_s} = \frac{9{,}55 \cdot P \cdot 16}{\pi \cdot D^3 \cdot n} \Rightarrow$$

$$= \frac{0{,}6R_e\pi \cdot n}{9{,}55 \cdot P \cdot 16} = \frac{1}{D^3} \Rightarrow D = \sqrt[3]{\frac{9{,}55 \cdot P \cdot 16}{0{,}6R_e\pi \cdot n}} = \sqrt[3]{\frac{9{,}55 \cdot 30000 \cdot 16}{0{,}6 \cdot 350 \cdot \pi \cdot 420}} =$$

$$= D = \sqrt[3]{\frac{9{,}55 \cdot 30000 \cdot 16}{0{,}6 \cdot 350 \cdot \pi \cdot 420}} = 2{,}55\,mm$$

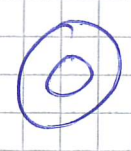
78 $D = 60\,mm$ $M_v = 6\,kNm$



$$W_v = \frac{\pi D^3}{16} = \frac{\pi \cdot 0{,}06^3}{16}$$

$$\tau = \frac{M_v}{W_v} = \frac{6}{4{,}24 \cdot 10^{-5}} = 141471\ N/mm^2 = 142\ N/mm^2$$

79



$$\frac{\pi(D^4 - d^4)}{16 \cdot D} = \frac{\pi(60^4 - 40^4)}{16 \cdot 60} = 34034$$

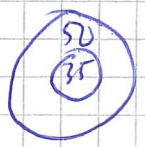
$$\tau = \frac{6000000}{34034} = 176\ N/mm^2$$

80 $M_v = 2{,}4\,kNm$ $d = 40\,mm$

$M_v = 2400000$ $W_v = \frac{\pi d^3}{16} = 12566$

$\tau = 198\ N/mm^2$

81



$M_v = 1{,}4\,kNm = 1400000\ Nmm$

$$W_v = \frac{\pi(50^4 - 35^4)}{16 \cdot 50} = 186501$$

$\tau = 75\ N/mm^2$

$$\frac{\pi \cdot 60^3}{16} = 42411$$

6000000

82 $D = 40\,mm$

$L = 300\,mm$

$R_e = 420\,N/mm^2$ $\quad n_s = 1,5$

$$\tau_{max} = 0,6\,\frac{R_e}{n_s} = 168\,N/mm^2$$

$$168 = \frac{F \cdot 300 \cdot 16}{\pi D^3} \Rightarrow F = \frac{168 \cdot \pi \cdot D^3}{300 \cdot 16} = 7037 = 7kN$$

83 $R_e = ?$ $\quad D = 30mm$ $\quad M_v = 400\,Nm$ $\quad n_s = 1,6$

$$\frac{0,6\,R_e}{1,6} = \frac{400000 \cdot 16}{\pi \cdot 30^3} \Rightarrow R_e = \frac{400000 \cdot 16 \cdot 1,6}{\pi \cdot 30^3 \cdot 0,6} = 201\,N/mm^2$$

84 $n_s = ?$ $\quad M_v = 900\,Nm$ $\quad R_e = 400\,N/mm^2$ $\quad d = 35mm$

$$\frac{0,6\,R_e}{n_s} = \frac{M_v}{W_v} \Rightarrow \frac{0,6 \cdot 400}{n_s} = \frac{900000 \cdot 16}{\pi \cdot 35^3} \Rightarrow$$

$$\Rightarrow n_s = \frac{0,6 \cdot 400 \cdot \pi \cdot 35^3}{900000 \cdot 16} = 2,24$$

85 $D = 30mm$ $\quad n = 600\,rpm$ $\quad R_e = 460\,N/mm^2$ $\quad n_s = 1,5$

$$\frac{0,6\,R_e}{n_s} = \frac{M_v}{W_v} \Rightarrow M_v = \frac{0,6 \cdot 460 \cdot \pi \cdot 30^3}{1,5 \cdot 16} = 97564\,Nmm = 975\,Nm$$

$P = M_v w$ $\quad w = \frac{2\pi \cdot 600}{60} = 62,8$

$$P = 975 \cdot 62,8 = 61\,kW$$

86 $R_e = ?$ $D = 25\,mm$ $P = 20000\,W$ $n = 480\,rpm$ $n_s = 1,4$

$$\frac{0,6R_e}{1,4} = \frac{397900 \cdot 16}{\pi \cdot 25^3} \Rightarrow$$

$$20000 = M_v \cdot 50,3 \Rightarrow M_v = 397,9\,Nm$$

$$w = \frac{2\pi \cdot 480}{60} = 50$$

$$\Rightarrow R_e = \frac{397900 \cdot 16 \cdot 1,4}{\pi \cdot 25^3 \cdot 0,6} = 302\,N/mm^2$$

87 $P = 30000\,W$ $n = 420\,rpm$ $R_e = 350\,N/mm^2$ $n_s = 2$

solut D

$$M_v = \frac{30000 \cdot 60}{2 \cdot \pi \cdot 420}$$

$$\frac{0,6 \cdot 350}{2} = \frac{30000 \cdot 60 \cdot 16}{2\pi \cdot 420 \cdot \pi \cdot D^3} \Rightarrow$$

$$D \Rightarrow \sqrt[3]{\frac{30000 \cdot 60 \cdot 16 \cdot 2}{0,6 \cdot 350 \cdot 2\pi \cdot 420 \cdot \pi}} = 3,2\,mm$$

$\sqrt[3]{1000}$

Se nästa sida

87

$$\frac{0,6R_e}{n_s} = \frac{9,55 \cdot P \cdot 16}{n \cdot \pi D^3}$$

$$D = \sqrt[3]{\frac{n_s \cdot 9,55 \cdot P \cdot 16}{0,6R_e \cdot n \cdot \pi}} = \sqrt[3]{\frac{2 \cdot 9,55 \cdot 30000 \cdot 16}{0,6 \cdot 350 \cdot 420 \cdot \pi}}$$

$$\frac{0,6R_e}{n_s} = \frac{\left(\frac{9,55 \cdot P}{n}\right)}{\left(\pi D^3/16\right)} = \frac{0,6R_e}{n_s} = \frac{9,55 \cdot P \cdot 16}{\pi \cdot D^3 \cdot n}$$

$$\frac{0,6R_e \cdot \pi \cdot n}{n_s \cdot 9,55 \cdot P \cdot 16} = \frac{1}{D^3} \Rightarrow D = \sqrt[3]{\frac{n_s \cdot 9,55 \cdot P \cdot 16}{0,6R_e \cdot \pi \cdot n}} =$$

=

87. $30kW = 30000W$ $\quad n = 420$

$R_e = 350$ $\quad n_s = 2$ $\quad$ Sökt d:

$$0{,}6 \cdot \frac{350}{2} = \frac{M_V}{W_V}$$

M_V $\quad \omega = \frac{2\pi}{60} \cdot 420 = 43{,}98$

$30000 = M_V \cdot 43{,}98 \Rightarrow M_V = 682\ Nm$

$$\frac{1}{W_V} = \frac{105}{M_V} \Rightarrow W_V = \frac{M_V}{105} = \frac{682000}{105} = 6{,}49 \cdot 10^3$$

$$6495 = \frac{\pi D^3}{16} \Rightarrow D = \sqrt[3]{\frac{6495 \cdot 16}{\pi}} = 32mm$$